

Venue Activity Prediction: a model for Social Media Integration

(working title)

Your name

October 10, 2026

Abstract

Placeholder. Write the abstract last, once the model and the simulation results exist.

1 Introduction

2 Literature Review

3 Problem Statement

A social app lets its users post a check-in at a public venue. Each check-in records one drink, so a single user can post more than once during one visit. Every check-in carries three things: one user, one timestamp, and one venue. The app accepts a check-in only from a user at the venue's own location, so a check-in cannot come from somewhere else.

The app serves a community of young adults who want to socialize with friends. We treat the community as a finite set of people, fixed only at the moment we judge it. Its size can differ from one night to the next. Deciding who belongs to that set in practice stays out of scope. We call a member who uses the app an *adopter*, and we call the percentage of the community who are adopters the *adoption rate*. The app also holds a social graph of the mutual follows between its adopters.

The app judges a venue from the check-ins posted there inside a window of two hours ending at the present moment. Within that window we see which adopters posted at the venue and how many posts each of them made. We see nothing of the members who stayed silent, and nothing of anyone outside the community.

Assumption 1 (No non-adopter signal). *Non-adopters post nothing.*

Assumption 2 (Adopter- and venue-dependent posting probability). *Every adopter present at a venue posts at least once with some probability. That probability can depend on the adopter and on the venue.*

Assumption 3 (Friends at the same venue and time). *Members who are not friends post independently of one another. Friends who are at the same venue at the same time do not post independently of each other.*

Without these three assumptions no procedure could recover anything from the check-in record, and the research question below would have no answer.

A user opens the app to decide where to go out, so the app must judge how much community activity each venue holds right now, and it reports that judgement coarsely. It shows every venue with at least one check-in inside the window, and it places that venue on one of three levels, which we label *quiet*, *busy*, and *packed*, shown respectively as yellow, orange, and red. Two boundaries separate the levels, set at the terciles of the occupancy distribution among currently

open venues, so each judged moment splits into equal thirds by that moment's own activity relative to each venue's own capacity, rather than by raw headcount or against a fixed historical count. A venue with no check-ins inside the window does not appear on the map. The user compares levels, not numbers.

Definition 1 (True headcount and occupancy). *At the present moment each venue holds a definite number of community members. We call that number the venue's true headcount. Each venue also has a fixed, known capacity. We call the venue's true headcount divided by its capacity its true occupancy. The app never reports a headcount or an occupancy and does not need to, because occupancy matters only for the level it falls in.*

Definition 2 (Match rate and coverage). *The app gets a shown venue right when it places that venue on the level its true occupancy falls in. Across the shown venues, we measure the fraction placed correctly and call it the match rate. A venue that never appears carries a different kind of error, so we measure it apart: among venues whose true occupancy reaches busy or packed, we measure the fraction that appear at all, and we call that the coverage.*

Our question concerns how the match rate and the coverage depend on adoption rate, holding four other quantities fixed: the number of venues, the number of members out, how unequally those members spread across venues, and how often an adopter posts. We ask for the smallest adoption rate at which the match rate reaches 80 percent and the coverage reaches 80 percent, and we ask whether such an adoption rate exists at all.

We put the match rate to every possible procedure rather than to one, because the app could compute its levels in more than one way and this statement names none of them. Coverage needs no such care, since a venue appears according to whether anyone posted and no procedure changes that.

Real check-in data exists for this app, but we do not use it here. This paper's aim is to find the minimal adoption rate any algorithm needs before the app implements it, and no implemented algorithm yet exists to test against that data. We keep real venue data out of scope as well. We also set aside what the map does to the people who read it, so every statement we make describes a world in which nobody saw the map.

4 Model

5 Simulation

6 Results

7 Discussion

References